



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2012
YEAR 11 Mathematics
Yearly

Mathematics Extension Continuers

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Answer must be given in simplest exact form.

Total Marks – 70

- Attempt questions 1-15

Examiner: *P. Bigelow*

Section I (10 marks)

Answer this section on the Multiple Choice Answer Sheet

(1) Evaluate $\lim_{a \rightarrow 4} \frac{a^2 - 16}{a + 4}$

(A) 0

(B) 8

(C) -4

(D) 4

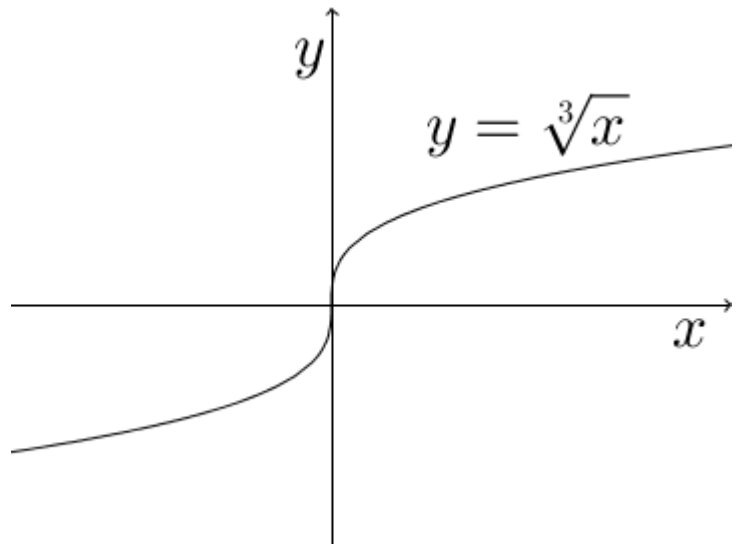
(2) At the origin:

(A) $y'' > 0$

(B) $y'' < 0$

(C) $y'' = 0$

(D) y'' is undefined



(3) A pair of dice are rolled, the probability the sum of the uppermost faces being greater than 7 is:

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) $\frac{5}{12}$

(D) $\frac{1}{3}$

(4) The full solution to $\frac{4}{x} > 1$ is:

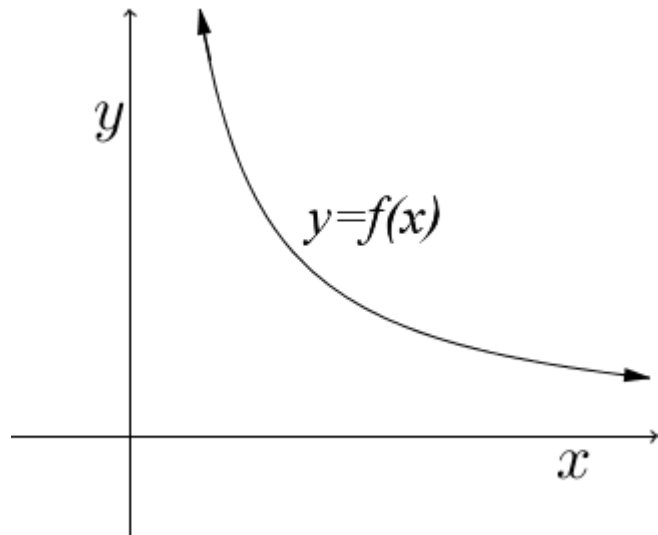
- (A) $x < 4$
- (B) $x > 4$
- (C) $0 < x < 4$
- (D) $x < 0, x > 4$

(5) Which of the following is not equal to $\cos 2\theta$

- (A) $\cos^2 \theta - \sin^2 \theta$
- (B) $1 - 2 \sin^2 \theta$
- (C) $2 \cos^2 \theta - 1$
- (D) $1 - 2 \cos^2 \theta$

(6) Which is true for $y = f(x)$?

- (A) $f'(x) < 0, f''(x) > 0$
- (B) $f'(x) < 0, f''(x) < 0$
- (C) $f'(x) > 0, f''(x) > 0$
- (D) $f'(x) > 0, f''(x) < 0$



(7) $3^x \times 2^x =$

- (A) 5^x
- (B) 5^{2x}
- (C) 6^{2x}
- (D) 6^x

(8) If $\log_5 x = 4$ then x is:

(A) 20

(B) 25

(C) 625

(D) 125

(9) A coin is tossed three times. The probability of getting 2 Heads and a Tail is

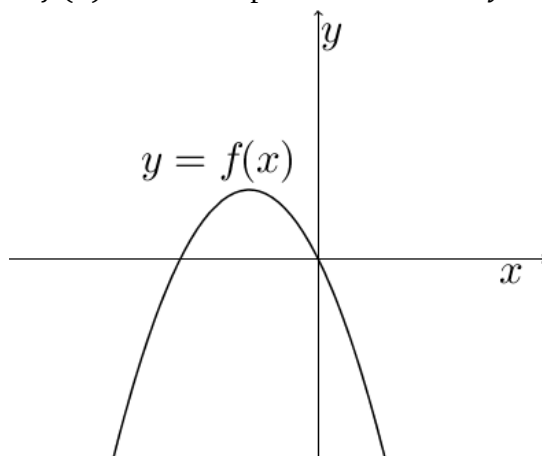
(A) $\frac{2}{3}$

(B) $\frac{1}{3}$

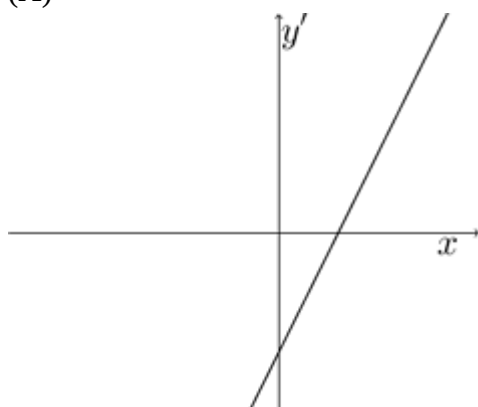
(C) $\frac{3}{8}$

(D) $\frac{5}{8}$

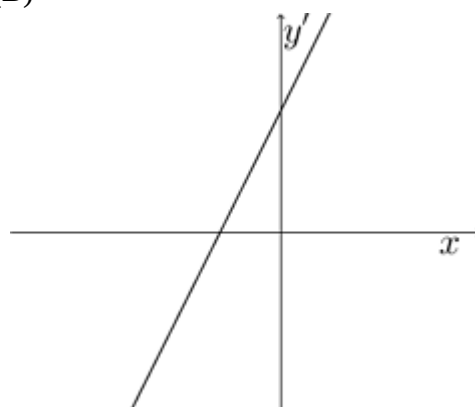
(10) Given the graph of $y = f(x)$ which is a possible sketch of $y = f'(x)$



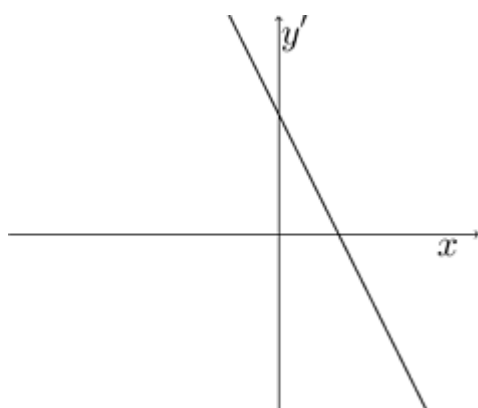
(A)



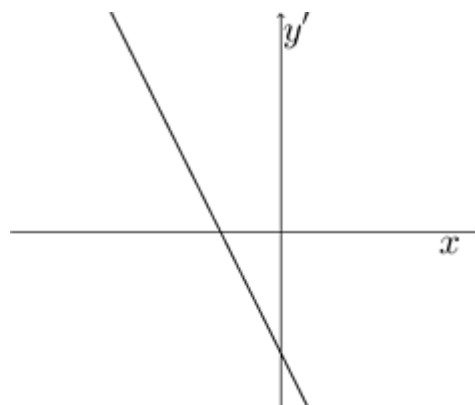
(B)



(C)



(D)



Section II (60 marks)

Answer each Question in a new Writing Booklet

Question 11 [12 marks]

(a) Simplify: $\sin 70^\circ \cos 40^\circ + \sin 40^\circ \cos 70^\circ$ [2]

(b) Expand $\cos(3A - 2B)$ [2]

(c) From an urn containing 4 white and 5 brown balls, two balls are selected without replacement. Find the probability of selecting [6]

(i) two white balls.

(ii) two different colours.

(iii) at least one brown.

(d) Simplify [2]

$$\frac{\sin 2A}{1 + \cos 2A}$$

End of Question 11

Answer each Question in a new Writing Booklet

Question 12 [12 marks]

(a) [3]

(i) Show that $\frac{\pi}{12} = \frac{\pi}{4} - \frac{\pi}{6}$

(ii) Hence find in simplest exact form the value of $\tan \frac{\pi}{12}$.

(b) Simplify [2]

(i) $\log_6 3 + \log_6 2$

(ii) $\log_5 100 - \log_5 4$

(c) Solve for x [2]

$$5^{3x-4} = 25^{x-2}$$

(d) If $f'(x) = 3x^2 + 2x + 4$ and $f(1) = 7$, find $f(-1)$. [3]

(e) If $f(x) = x^2 + 3x$ find $f'(x)$ from first principles. [2]

End of Question 12

Answer each Question in a new Writing Booklet

Question 13 [12 marks]

- (a) Solve the following inequations and plot the solutions on separate number lines.

[4]

(i) $\frac{3}{|2x-1|} > 1$

(ii) $\frac{4}{2x-1} \leq \frac{1}{x+2}$

- (b) Find $\frac{dy}{dx}$ for the following

[8]

(i) $y = \sqrt{7+x^2}$

(ii) $y = x\sqrt{2x-1}$

(iii) $y = \sqrt[3]{x^2} + \sqrt{x^3}$

(iv) $y = \frac{5x}{4+x^2}$

End of Question 13

Answer each Question in a new Writing Booklet

Question 14 [12 marks]

- (a) Find [2]

$$\lim_{x \rightarrow \infty} \frac{4x + 3x^2}{5x - 2x^2}$$

- (b) If [2]

$$f(x) = x^2 + \frac{1}{x^2}$$

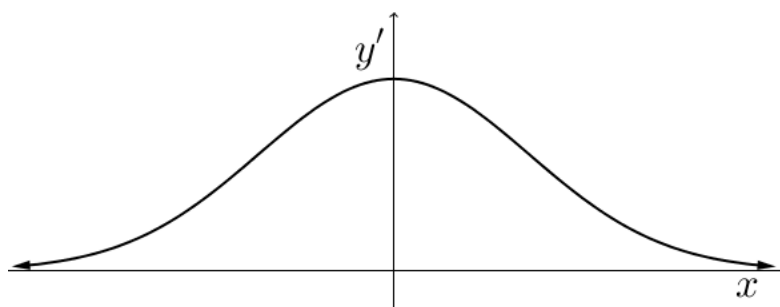
find

(i) $f'(x)$

(ii) $f''(x)$

- (c) For what values of x is $f(x) = x^3 - 3x^2 - 7x + 10$ concave up? [2]

- (d) Given that the graph below is the gradient function of $y = f(x)$.
Sketch $y = f(x)$. [2]



- (e) If $\tan A = \frac{3}{4}$ and $\cos B = \frac{5}{13}$, where A and B are acute. Find: [4]

(i) $\sin 2A$.

(ii) $\cos(A - B)$.

End of Question 14

Answer each Question in a new Writing Booklet

Question 15 [12 marks]

(a)

[4]

(i) If $\theta + \varphi = 45^\circ$ show that

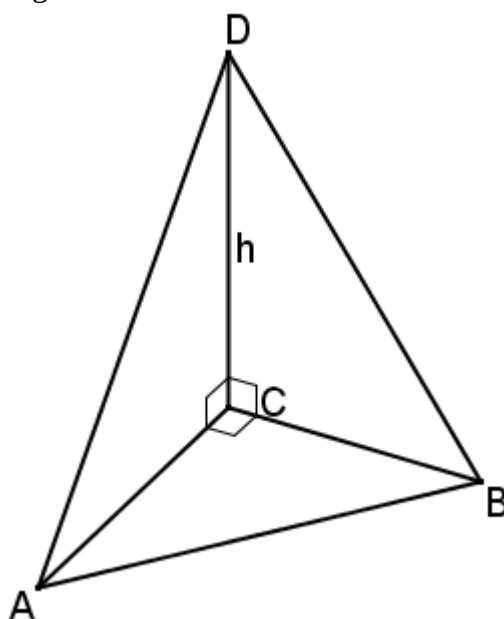
$$\tan \theta + \tan \varphi = 1 - \tan \theta \tan \varphi$$

(ii) By letting $\theta = \varphi$ show that $t^2 + 2t - 1 = 0$ where $t = \tan 22\frac{1}{2}^\circ$.

(iii) Hence find the exact value of $\tan 22\frac{1}{2}^\circ$.

(b) A tower CD is of height h metres. From a point A due South of the base the angle of elevation of D is 32° . From B, due East of the base the angle of elevation of D is 28° . Given that A and B are 500 metres apart: find the height h to the nearest metre.

[4]

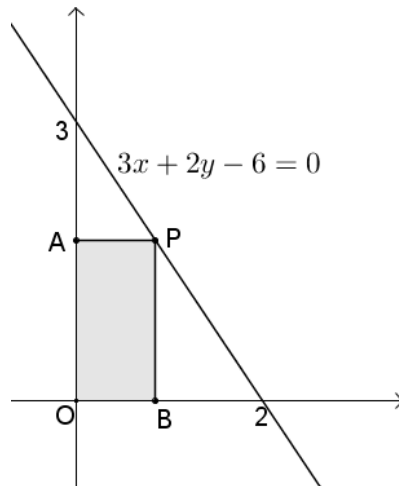


- (c) Two sides of a rectangle OAPB lie on the x and y axes. The vertex opposite the origin lies in the first quadrant and is on

[4]

$$3x + 2y - 6 = 0$$

Find the maximum area of the rectangle.



End of Exam



Student Number: _____

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Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Section I: Multiple choice answer sheet.

Completely colour the cell representing your answer. Use black pen.

1. ☒ A ☐ B ☐ C ☐ D
2. ☐ A ☐ B ☐ C ☒ D
3. ☐ A ☐ B ☒ C ☐ D
4. ☐ A ☐ B ☒ C ☐ D
5. ☐ A ☐ B ☐ C ☒ D
6. ☒ A ☐ B ☐ C ☐ D
7. ☐ A ☐ B ☐ C ☒ D
8. ☐ A ☐ B ☒ C ☐ D
9. ☐ A ☐ B ☒ C ☐ D
10. ☐ A ☐ B ☐ C ☒ D

$$\begin{aligned} \text{a) } \sin 70^\circ \cos 40^\circ + \sin 40^\circ \cos 70^\circ \\ &= \sin(70^\circ + 40^\circ) \quad (2) \\ &= \underline{\underline{\sin(110^\circ)}} \end{aligned}$$

$$\begin{aligned} \text{b) } \cos(3A - 2B) \\ &= \underline{\cos 3A \cos 2B} \quad (2) \\ &\quad + \underline{\sin 3A \sin 2B} \\ &= 1\frac{1}{2} \end{aligned}$$

c) 4W 5B

i) $P(2 \text{ white}) = \frac{1}{6} \quad (2)$

$$\begin{aligned} \text{ii) } P(2 \text{ diff}) &= P(WB) + P(BW) \\ &= \left(\frac{4}{9} \times \frac{5}{8}\right) + \left(\frac{5}{9} \times \frac{4}{8}\right) \\ &= \frac{5}{9} \quad (2) \end{aligned}$$

$$\begin{aligned} \text{iii) } P(\text{at least 1 brown}) \\ &= 1 - P(\text{no brown}) \\ &= 1 - \frac{1}{6} \\ &= \frac{5}{6} \quad (2) \end{aligned}$$

$$\text{d) } \frac{\sin 2A}{1 + \cos 2A}$$

$$= \frac{2 \sin A \cos A}{1 + [2 \cos^2 A - 1]}$$

$$= \frac{2 \sin A \cancel{\cos A}}{2 \cancel{\cos^2 A}}$$

$$= \frac{\cancel{2} \sin A}{\cancel{2} \cos A}$$

$$= \underline{\underline{\tan A}} \quad (2)$$

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(12) (a) (i) show $\frac{\pi}{12} = \frac{\pi}{4} - \frac{\pi}{6}$

RHS $\frac{\pi}{4} - \frac{\pi}{6} = \frac{3\pi - 2\pi}{12} = \frac{\pi}{12}$ (1)

(ii) $\tan \frac{\pi}{12} = \tan \left(\frac{\pi}{4} - \frac{\pi}{6} \right)$
 $= \tan \frac{\pi}{4} - \tan \frac{\pi}{6}$

$$\frac{1 - \tan \frac{\pi}{4} \tan \frac{\pi}{6}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{6}}$$
$$= \frac{1 - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1}$$

$$= \frac{\sqrt{3}-1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)}{(\sqrt{3}+1)} \text{ rationalized}$$

$$\frac{(\sqrt{3}-1)}{(\sqrt{3}+1)} \times \frac{(\sqrt{3}-1)}{(\sqrt{3}-1)} = \frac{3-2\sqrt{3}+1}{3-1} = \frac{4-2\sqrt{3}}{2}$$

$$= \frac{2(2-\sqrt{3})}{2}$$

$$= 2 - \sqrt{3} \quad (2)$$

(12) (b) (i) $\log_6 3 + \log_6 2 = \log_6 (3 \times 2)$
 $= \log_6 6 = 1$ (1)

(ii) $\log_5 100 - \log_5 4 = \log_5 \left(\frac{100}{4} \right)$
 $= \log_5 25$
 $= \log_5 5^2 = 2 \log_5 5 = 2$ (1)

(12) (c) $5^{3x-4} = 25^{x-2}$
 $5^{3x-4} = 5^{2(x-2)} \Rightarrow 3x-4 = 2x-4$
 $x = 0$ (2)

(d) $f'(x) = 3x^2 + 2x + 4$
 $f(x) = \frac{3x^3}{3} + \frac{2x^2}{2} + 4x + C$
 $f(x) = x^3 + x^2 + 4x + C$

$f(1) = 7$
 $7 = 1 + 1 + 4 + C$
 $C = 1$

$f(x) = x^3 + x^2 + 4x + 1$

So $f(-1) = -1 + 1 - 4 + 1 = -3$ (3)

(e) $f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 + 3(x+h) - (x^2 + 3x)}{h}$
 $\lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 3x + 3h - x^2 - 3x}{h}$
 $\lim_{h \rightarrow 0} \frac{2xh + h^2 + 3h}{h}$
 $\lim_{h \rightarrow 0} (2x + h + 3)$

$f'(x) = 2x + 3$

(2)

(13) (a) (i) $\frac{3}{|2x-1|} > 1 \quad x \neq \frac{1}{2}$

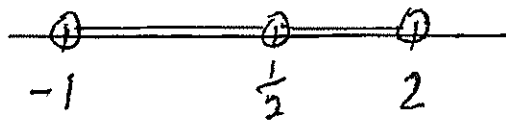
$$3 > |2x-1|$$

$$|2x-1| < 3$$

$$2x-1 < 3 \quad 2x-1 > -3$$

$$2x < 4 \quad 2x > -2$$

$$x < 2 \quad x > -1$$



(2)

(ii) $\frac{4}{(2x-1)} \leq \frac{1}{(x+2)} \quad x \neq \frac{1}{2}, x \neq -2$

$$x(2x-1)^2(x+2)^2 \quad (2x-1)^2(x+2)^2 \times \frac{4}{(2x-1)} \leq \frac{1}{(x+2)} \times (2x-1)^2(x+2)^2$$

$$4(2x-1)(x+2)^2 \leq (2x-1)^2(x+2)$$

$$4(2x-1)(x+2)^2 - (2x-1)^2(x+2) \leq 0$$

$$(2x-1)(x+2) [4(x+2) - (2x-1)] \leq 0$$

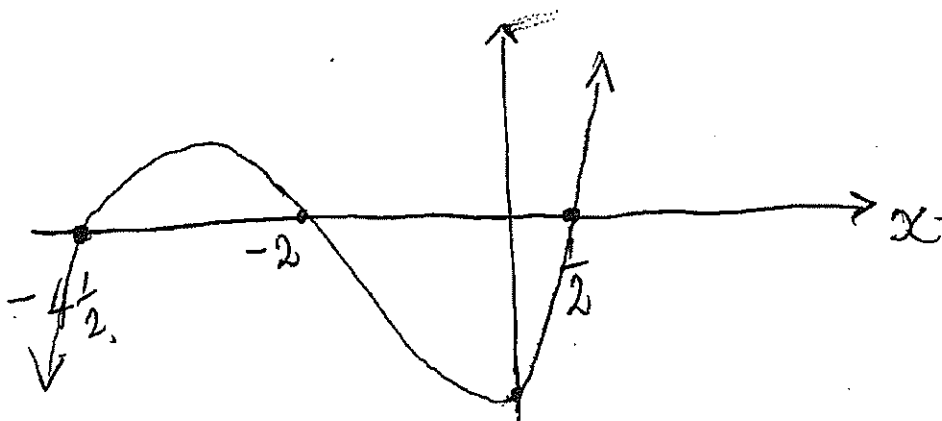
$$(2x-1)(x+2) [4x+8-2x+1] \leq 0$$

$$(2x-1)(x+2) [2x+9] \leq 0$$

below or touching

$x \leq -4\frac{1}{2}$, $-2 < x < \frac{1}{2}$
with restriction

(2)



$$\begin{aligned}
 (13) \quad (b) \quad (i) \quad y &= \sqrt{7+x^2} \\
 &= (7+x^2)^{\frac{1}{2}} \\
 y' &= \frac{1}{2} (7+x^2)^{-\frac{1}{2}} \times 2x \\
 &= \frac{x}{\sqrt{7+x^2}} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad y &= x \cdot \sqrt{2x-1} \\
 &= x (2x-1)^{\frac{1}{2}} \\
 y' &= x \times \frac{1}{2} (2x-1)^{-\frac{1}{2}} \times 2 + (2x-1)^{\frac{1}{2}} \times 1 \\
 &= \frac{x}{\sqrt{2x-1}} + \frac{\sqrt{2x-1}}{1} = \frac{x + 2x-1}{\sqrt{2x-1}} = \frac{3x-1}{\sqrt{2x-1}} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad y &= \sqrt[3]{x^2} + \sqrt{x^3} \\
 &= x^{\frac{2}{3}} + x^{\frac{3}{2}} \\
 y' &= \frac{2}{3} x^{-\frac{1}{3}} + \frac{3}{2} x^{\frac{1}{2}} \\
 &= \frac{2}{3\sqrt[3]{x}} + \frac{3\sqrt{x}}{2} \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad y &= \frac{5x}{4+x^2} \\
 y' &= \frac{(4+x^2) \times 5 - 5x \times 2x}{(4+x^2)^2} \quad (2) \\
 &= \frac{20+5x^2-10x^2}{(4+x^2)^2} = \frac{20-5x^2}{(4+x^2)^2} \text{ or } \frac{5(4-x^2)}{(4+x^2)^2}
 \end{aligned}$$

14(a)

$$\lim_{x \rightarrow \infty} \frac{4x + 3x^2}{5x - 2x^2} = \lim_{x \rightarrow \infty} \left(\frac{\frac{4}{x} + 3}{\frac{5}{x} - 2} \right)$$

$$= -\frac{3}{2}$$

(b)

$$f(x) = x^2 + x^{-2}$$

$$f'(x) = 2x - 2x^{-3} = 2x - \frac{2}{x^3}$$

$$f''(x) = 2 + 6x^{-4} = 2 + \frac{6}{x^4}$$

(c)

$f(x)$ concave up when $f''(x) > 0$

$$f(x) = x^3 - 3x^2 - 7x + 10$$

$$f'(x) = 3x^2 - 6x - 7$$

$$f''(x) = 6x - 6$$

Concave up $6x - 6 > 0$

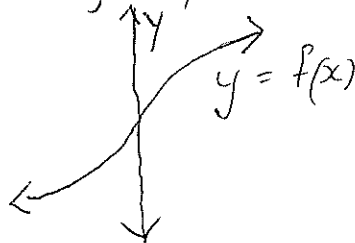
$$x > 1$$

(d)

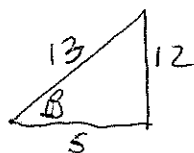
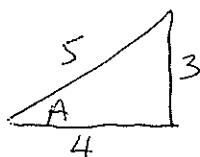
$y' > 0$ y increasing

y' increasing y concave up

y' decreasing y concave down



(e)



$$\sin 2A = 2 \sin A \cos A = 2 \times \frac{3}{5} \times \frac{4}{5} = \frac{24}{25}$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$= \frac{4}{5} \times \frac{5}{13} + \frac{3}{5} \times \frac{12}{13}$$

$$= \frac{56}{65}$$

$$15(i)(i) \quad \tan(\theta + \alpha) = \frac{\tan\theta + \tan\alpha}{1 - \tan\theta \tan\alpha}$$

$$\tan 45^\circ = 1 = \frac{\tan\theta + \tan\alpha}{1 - \tan\theta \tan\alpha}$$

$$\therefore \tan\theta + \tan\alpha = 1 - \tan\theta \tan\alpha$$

$$(ii) \quad \tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

$$2\theta = 45^\circ \quad \theta = 22\frac{1}{2}^\circ \quad t = \tan 22\frac{1}{2}^\circ$$

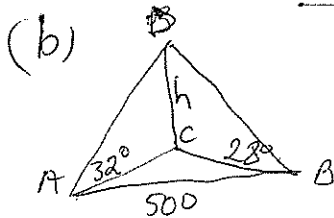
$$1 = \frac{2t}{1 - t^2}$$

$$2t = 1 - t^2$$

$$t^2 + 2t - 1 = 0$$

$$(iii) \quad \text{from (ii)} \quad t = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times -1}}{2}$$

$$t = -1 \pm \sqrt{2} \quad t = \sqrt{2} - 1 \quad (t > 0)$$



$$\tan 32^\circ = \frac{h}{AC} \quad \tan 28^\circ = \frac{h}{BC}$$

$$AC^2 + BC^2 = 500^2$$

$$h^2 \cot^2 32^\circ + h^2 \cot^2 28^\circ = 500^2$$

$$h^2 = \frac{500^2}{\cot^2 32^\circ + \cot^2 28^\circ}$$

$$h = 202 \text{ m nearest m}$$

$$(c) \quad A = xy, \quad 2y = 6 - 3x$$

$$y = 3 - \frac{3}{2}x$$

$$A = 3x - \frac{3}{2}x^2$$

$$\frac{dA}{dx} = 3 - 3x$$

$$\text{max } 3 - 3x = 0$$

$$x = 1, \quad y = \frac{3}{2}$$

$$\text{Max } A = \frac{3}{2} \text{ u}^2$$
